TRƯỜNG ĐẠI HỌC BÁCH KHOA HÀ NỘI **TRƯỜNG ĐIỆN – ĐIỆN TỬ** **Number: 02** **No. pages: 4**	**MIDTERM EXAM 2022.1** **COURSE ID: ET3300E – DIGITAL SYSTEM DESIGN 2** **ELECTRONIC & TELECOM ENGINEERING** **Date: 29/12/2022** **Time duration: 90 minutes** *(Books are allowed, PCs are allowed without using keyboard. Return these sheets with answers filled)*
Signatures Instructor:	Group leader:
Student name:	**Student ID:**
Total points: 240	**You get: /240**

Note that:

- For questions with required explanation, the right choice(s) of your answer is only made 40% of points. The rest of 60% of points is made by details you provide.

PART A: QUESTIONS FOR THE INTEL MICROPROCESSORS				
Question 1. What is the instruction moving data from a memory location to a register?			**Point: 10**	**Get:**
A. MOV CH, [DI+1234h]	B. MOV [DI+1234h], CH	C. MOV AL, 1234h	D. MOV DL, [SI+1234H]	
For your choice(s), please give brief explanation for the data size, and what is the physical address of memory location? …… …… ……				
Question 2. What is the instruction moving data from a register to a memory location?			**Point: 10**	**Get:**
A. MOV AH, [1234h]	B. MOV [1234h], DX	C. MOV [SI+34h], BX	D. MOV AX, [BX+SI]	
For your choice(s), please give brief explanation the data size, and physical address of memory location? …… …… ……				

Question 3. What is the meaning/activity of the following instructions in the 8086?		Point: 50	Get:
No.	**Instruction**	**Detailed Activity/meaning**	
1.	MOV WORD PTR [SI], [DI]		
2.	MOV CX, [SI+1234h]		
3.	PUSH BX		
4.	POP AX		
5.	JMP BX		
6.	JNZ LabelB		
7.	MOVSW		
8.	ROL CL,04		
9.	IN AL, 8Fh		
10.	OUT DX, AL		
11.	TEST AX, CX		
12.	CMP BL,02		
13.	SHL CL,10		
14.	DIV DL		
15.	MUL BL		
16.	IDIV CX		
17.	IMUL DL		
18.	LOOP NEXT		
19.	MOVSB		
20.	INT 21h		

Question 4. What are the machine codes of following instructions?		Point: 50	Get:
No.	**Instruction**	**Machine languages/codes**	
1.	ADD BH, CL		

2.	SUB DH, CH	
3.	CMP CL,02h	
4.	DIV DH	
5.	MUL CL	
6.	STI	
7.	PUSH BX	
8.	MOV BH,01h	
9.	IN AL, 0Fh	
10.	OUT 0Fh, AX	

Question 5. Please draw the block diagrams of Von Neuman architecture and Harvard architecture?	**Point: 30**	**Get:**

Please give details about them.

……

……

……

……

……

……

……

……

……

……

……

……

……

……

……

……

……

……

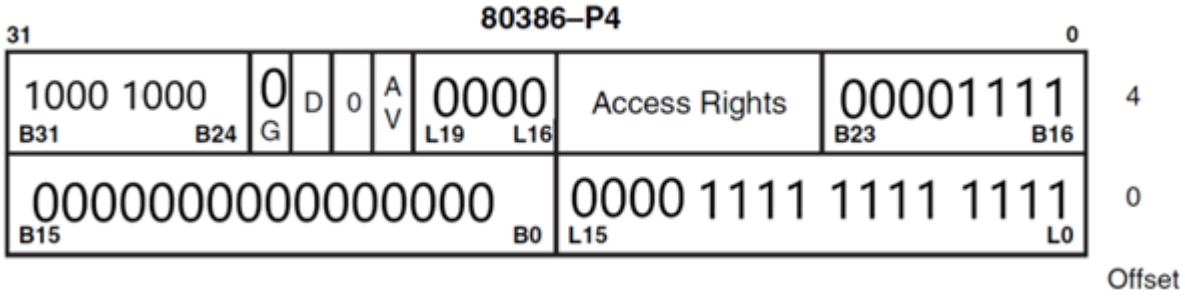
……

……
……
……
……
……
……
……
……
……
……
……

Question 6. Please calculate physical addresses in the real mode for 8086/88?	**Point: 20**	**Get:**

1.	0321:1230h	
2.	0320:1231h	
3.	F001:FF00h	
4.	F000:FF01h	
5.	B000:BBC8h	

Question 7. Give a descriptor of 80386 as below



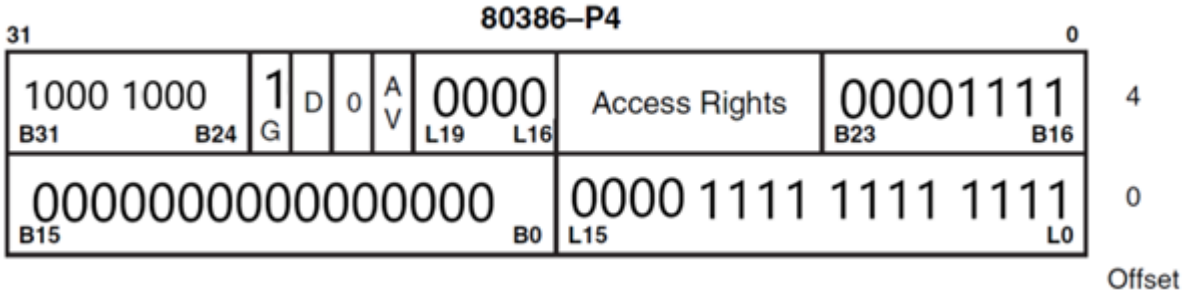
Please answer the below questions by filling the number or circle:

1. What is the range of address of the segment with above descriptor?	**Point: 10**	**Get:**
The range of address in hexadecimal is from …….. to.. ……		

2. The memory location at 880F0FF8H is belonged to the segment described by above descriptor?			**Point: 10**	**Get:**
A. NO	B. YES	C. MAYBE		

3. The memory location at 881F0FFEh is belonged to the segment described by above descriptor?			**Point: 10**	**Get:**
A. NO	B. YES	C. MAYBE		

Question 8. Give a descriptor of 80386 as below



Please answer the below questions by filling the number or circle:

1. What is the range of address of the segment with above descriptor?			**Point: 10**	**Get:**
The range of address in hexadecimal is from …….. to.. ……				
2. The memory location at 890EFFFFH is belonged to the segment described by above descriptor?			**Point: 10**	**Get:**
A. NO	B. YES	C. MAYBE		
3. The memory location at 9000 0000h is belonged to the segment described by above descriptor?			**Point: 10**	**Get:**
A. NO	B. YES	C. MAYBE		
Question 9. In protected mode, what is the content of segment register?			**Point: 10**	**Get:**
A. Segment address	B. indicate the index of chosen descriptor	C. Both A) and B)		

Please give the brief explanation (if any)

……

……

……

……